MUTHUKUMARAN YOGEESWARAN

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EXPERIENCE

Research Student, NUS Immersive Reality Lab **Aug 2025 – May 2026**

- Built a **fully offline conversational companion robot** for eldercare with **NUS Nursing**, with the whole voice and perception stack on one **NVIDIA Jetson**.
- Combined a speech pipeline (VAD, ASR, **on-device LLM**, TTS) with **YOLO pose estimation** and emotion recognition, so the robot senses user affect and adapts in real time.
- Designed and **3D-printed** a custom **2-DOF robotic head** across multiple prototypes, with **affect-driven facial expressions** lip-synced to speech.

Robot Operator, Cortex AI **Nov 2025 – Jan 2026**

- Collected training data for **Vision-Language-Action (VLA) models** through two pipelines: teleoperated **YAM robot arms** for manipulation demonstrations and **egocentric data capture with a Meta Quest 3**, designing new environments and tasks to expand dataset coverage.
- Worked hands-on with the **LeRobot** dataset framework and codebase, developing a strong intuition for evaluating **data quality** and distinguishing usable, high-signal demonstrations from low-quality or noisy data.

Research Intern, The University of Osaka **May 2025 – Aug 2025**

- Built a real-time **human localization system** for a social robot: trained a **custom YOLO keypoint model** and fused its detections into the robot's **SLAM map** via Kabsch alignment, reaching **0.02–0.06 m accuracy**.
- **Deployed live at Osaka Expo 2025** (Yoshikawa Lab pavilion), localizing visitors in crowds so the robot could meet their gaze.

Research Intern, A*STAR (Advanced Remanufacturing & Technology Centre) **May 2024 – Aug 2024**

- Trained a **dexterous robotic hand** to grasp and relocate objects in **MuJoCo physics simulation** using **DAPG**, a reinforcement learning method bootstrapped from human demonstrations.
- Adapted the entire **simulation pipeline** from a research hand (Adroit) to the 6-actuator Inspire hand, converting the robot model from URDF to MJCF and remapping the observation space.

Software Team Member, NUS RoboMaster **Aug 2023 – Dec 2024**

- Collaborated in a team of 4 to build a **two-legged self-balancing robot** for the DJI RoboMaster Championships, **simulating the robot dynamics and tuning LQR/PID control in Simulink** before deploying to **STM32**.
- Contributed to the chassis **CAD design** and prototyped a **computer vision aim-bot** for competitive auto-targeting.

PROJECTS

BoxBunny: AI Boxing Sparring & Coaching Robot (Final-Year Capstone) **Aug 2025 – May 2026**

- Led robot intelligence for an adaptive boxing robot, achieving **96.8% punch recognition at 30 fps** on unseen users with a custom **transformer model** optimized with **TensorRT** on a **Jetson Orin NX**.
- Architected the **ROS 2** runtime connecting perception, action recognition, and an **LLM-based coaching loop**, hardened with unit and integration testing.
- Curated a custom **RGB-D action dataset** with an **Intel RealSense D435i** and built an in-house annotation GUI for labelling and evaluation.
- Presented the project at the **International Senior Project Conference 2026** (KMUTT, Bangkok) and demoed it live at the NUS CDE AI Fair.

Breaking Bias: Bias Evaluation & Mitigation in LLMs (University of Toronto exchange) **Jan 2024 – May 2024**

- Investigated how AI chatbots reproduce gender, race, and religion stereotypes: benchmarked **Llama 2** and **GPT-3.5** on standard bias tests (CrowS-Pairs, StereoSet) using **PyTorch**, then tested whether **prompt engineering** and **fine-tuning** reduce the bias.

TECHNICAL SKILLS AND PROFESSIONAL CERTIFICATIONS

- **Programming Languages: Python**, **C++**, C, MATLAB, JavaScript, Assembly
- **Machine Learning & AI: PyTorch**, **Computer Vision (YOLO)**, Neural Networks, Action Recognition (Pose / Temporal Models), **Reinforcement Learning**, **Large Language Models (LLMs) + RAG**, TensorRT
- **Robotics & Systems: ROS 2**, **SLAM**, Robot Perception, **Teleoperation (LeRobot)**, Mobile Robotics, **NVIDIA Jetson**, Intel RealSense, STM32
- **Simulation & Control: MuJoCo**, MATLAB / **Simulink**, **PID / LQR Control**, COMSOL Multiphysics, SolidWorks Simulation
- **CAD & Design:** SolidWorks (**CSWA**, Mechanical Design & Additive Manufacturing)
- **Tools & Platforms:** Git, GitHub, **Docker**, Conda, **Linux**

EDUCATION

National University of Singapore, Bachelor of Engineering **Aug 2022 – Jul 2026**

- Major in Engineering Science with a specialisation in **Computational Engineering Science**
- Minors in Innovation & Design Programme and Computing (Design and Engineering)
- Focus areas: **Robotics, Artificial Intelligence, Human–Robot Interaction**

EXTRA CURRICULAR ACTIVITIES

President, NUS Engineering Science Student Wing **Aug 2023 – Aug 2024**

- Led a **40-member team**; planned and hosted **MakerMania23**, a national prototyping competition with roughly 40 teams from Junior Colleges, Polytechnics, and Universities.

NUS Student Ambassador, National University of Singapore **Jan 2023 – May 2026**

- Represented NUS in communications, outreach, and official university events.